From Drought to Deluge: Modeling Climate Volatility and Livability Risk in Southern Europe

Project Summary

A comparative climate risk analysis of two southern European cities — **Seville, Spain** (heat and wildfire exposure) and **Larissa, Greece** (drought-to-flood volatility in the Thessaly agricultural plain) — examining how climate stress indicators evolved from 1990–2020 and are projected to change through 2045 under a high-emissions scenario (SSP5-8.5). The project uses ERA5 reanalysis data, CMIP6/pre-built Copernicus climate indicators, and a custom volatility metric built from precipitation extremes.

Why these two cities: They represent two distinct climate "failure modes" rather than a single generic-warming narrative — Seville shows sustained, escalating heat stress, while Larissa demonstrates that the more dangerous emerging pattern in parts of southern Europe may be *swings between extremes* (severe drought followed by catastrophic flooding, as seen with Storm Daniel in September 2023) rather than steady drying alone.

Hypotheses to Test

1. **H1 — Heat escalation (Seville):** Heatwave frequency (≥3 consecutive days above the 90th-percentile historical threshold) and tropical nights (min temp ≥20°C) show a statistically significant upward trend 1990–2020, continuing/accelerating under SSP5-8.5 through 2045.

2. **H2 — Compound fire risk (Seville):** Fire danger index values correlate more strongly with combined heat+dryness conditions than with temperature alone — wildfire risk is better predicted by a compound indicator than by heat data alone.

3. **H3 — Drought severity (Larissa):** The 3-month SPI series for Thessaly shows increasing frequency and/or severity of drought values over the historical record.

4. **H4 — Volatility, not just drying (Larissa — central hypothesis):** Climate instability in Thessaly — measured as severe drought followed within the same year by extreme precipitation — is increasing in frequency, meaning the region's central risk is *swing magnitude* rather than a simple drying trend. Directly testable against the documented 2023 drought→Storm Daniel sequence.

5. **H5 — Divergent regional risk (cross-city):** The two cities' climate stress profiles diverge in *character*, not just magnitude — southern Europe's climate risk

is geographically heterogeneous, not a single uniform "getting worse" story.

H4 and H5 are the strongest, most original contributions — they go beyond restating known warming trends into a specific, testable claim about volatility and regional heterogeneity.

Datasets

Purpose	Dataset	Cities	Period
Historical baseline (temp, precip, wind, RH)	`reanalysis-era5-single-levels` (CDS)	Both	1990–2020
Pre-built climate indicators (check first — may cover heatwave/tropical night calcs already)	`sis-ecde-climate-indicators` (CDS)	Both	1940–2100
Raw projections (fallback if pre-built indicators don't cover it)	`projections-cmip6`, SSP5-8.5 scenario	Both	2025–2045
Fire danger, historical	CEMS Fire danger historical dataset (CDS)	Seville	1990–2020
Fire danger, projected	"Fire danger indicators for Europe" (RCP4.5/8.5, FWI-based)	Seville	2025–2045
Precipitation for SPI	Same ERA5 pull as baseline (`total_precipitation` variable); optional E-OBS cross-check	Larissa	1990–2020

Scope note for the write-up: the fire danger projections use RCP scenarios (CMIP5-era), not SSP5-8.5 (CMIP6-era) — roughly comparable in severity but not technically the same framework. Worth one sentence in the limitations section.

Main Tasks

Week 1 — Setup + historical data
Register CDS access; pull ERA5 historical data for both cities; check whether `sis-ecde-climate-indicators` already covers needed indicators; load and sanity-check raw time series.

Week 2 — Historical indicators + validation
Compute Seville heatwave-day and tropical-night counts; pull/compute fire danger index; compute Larissa 3-month SPI using `standard_precip` or `spei`; validate against known events (2022 Seville heat, 2023 Larissa drought→flood) — do not proceed until these show up clearly in your own computed data.

Week 3 — Projections + trend analysis
Pull SSP5-8.5 projection data (or use pre-built indicators through 2045); apply the same indicator logic to projected data using the same historical baseline thresholds; fit simple linear trends (not advanced time series models); produce before/after comparison charts for each indicator.

Week 4 — Write-up + polish
Write the comparative narrative (steady escalation vs. volatility); add brief Po Valley/Évora context paragraphs from published sources (no pipeline needed);

write
an explicit limitations section; polish chart labels/titles; attempt stretch goals only
if ahead of schedule.

Expected / Potential Results

- **Seville:** A visible, likely statistically significant upward trend in both heatwave
  frequency and tropical-night counts across the historical record, with the trend
  continuing (and plausibly steepening) in the 2025–2045 projection — consistent with
  published findings that southern Europe, and Iberia specifically, is warming faster
  than the global average and is already setting individual national heat records.
- **Seville fire risk:** Fire danger index values that track more closely with combined
  heat-and-dryness periods than temperature alone, supporting H2 and giving you a
  concrete compound-risk chart rather than a single-variable one.
- **Larissa:** A historical SPI series with a clear, sharp negative dip in the months
  preceding September 2023, validating the pipeline, alongside a broader pattern of
  increasingly large swings between SPI minimums and maximum short-window precipitation
  totals — the core evidence for the volatility hypothesis (H4).
- **Cross-city comparison:** Two visibly different risk "shapes" when plotted side by
  side — a steady upward slope for Seville vs. a widening oscillation band for Larissa —
  which is the visual anchor for H5 and the centerpiece figure of the final report.
- **Honest caveat to state upfront:** with two cities, one model, and one scenario, these
  results are illustrative of directional risk rather than definitive climate-model
  consensus — appropriate framing for a capstone, and worth saying explicitly rather than
  overclaiming.

What Got Cut From the Original Scope (and why)
- 4 cities → 2 cities (debugging surface roughly doubles per added city)
- 2 emissions scenarios → 1 (SSP5-8.5 only)
- Custom-built Fire Weather Index → pre-built Copernicus fire danger data
- Composite cross-city index → stretch goal only, attempted last if time allows
- Advanced time series modeling (e.g. ARIMA) → simple linear trend fitting